MATH 205

Linear Algebra

LECTURE 4 · GAUSS-JORDAN ELIMINATION & THE BIG PICTURE

From reduced row-echelon form to the geometry of Ax = b — free variables, parametric solutions, and what a system of lines can tell you about a matrix.

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WELCOME BACK

Where Today Fits

Last time

- Computing A^{-1} via $[A \mid I] \rightarrow [I \mid A^{-1}]$, and why the method works
- Row-echelon form & Gaussian elimination, finished with back-substitution

Today

- From echelon form to reduced row-echelon form — Gauss-Jordan elimination
- Free variables and the parametric form of a solution set
- The geometry of $Ax = b$: intersecting, parallel, and coincident lines
- Revisiting A^{-1} for 2×2 matrices, and the determinant

Course Roadmap

1. Systems of Linear Equations
2. **Matrices & Matrix Algebra**
3. Determinants
4. Vector Spaces
5. Eigenvalues & Eigenvectors

Row reduction is now a complete toolkit — next time, the many faces of invertibility come together in one theorem.

1.3 REDUCED ROW-ECHELON FORM

Finishing What Echelon Form Started

Gaussian elimination gets you to echelon form, then back-substitution finishes the job.

Push the row reduction one step further — clear above every pivot too — and the system solves itself.

Reduced Row-Echelon Form

A matrix is in reduced row-echelon form (RREF) when it satisfies two conditions:

1. It is already in row-echelon form.
2. Each column that contains a leading 1 has zeros everywhere else — above the pivot as well as below it.

Are both matrices below in RREF? Trace each leading 1 and check its column is otherwise all zero:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

1.3 WORKED EXAMPLE

Solve by Gauss-Jordan Elimination

Solve the system by reducing the augmented matrix all the way to reduced row-echelon form:

$$\begin{cases} 3x_1 + 4x_2 + 5x_3 = 12 \\ x_1 - x_2 + 2x_3 = 2 \\ 2x_1 + x_2 + 3x_3 = 6 \end{cases}$$

Augmented matrix

$$\left[\begin{array}{ccc|c} 3 & 4 & 5 & 12 \\ 1 & -1 & 2 & 2 \\ 2 & 1 & 3 & 6 \end{array}\right]$$

→

Echelon form (Lecture 3)

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 2 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

Ordinary Gaussian elimination would stop here and back-substitute for x_3, x_2, then x_1. Gauss-Jordan keeps row-reducing instead — clearing above the pivots as well as below them.

Clearing Above the Pivots

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 2 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 3 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$R_1 \rightarrow R_1 + R_2$

$R_1 \rightarrow R_1 - 3R_3,\ R_2 \rightarrow R_2 - R_3$

Two more replacement steps clear column 3 out of rows 1 and 2, then column 1 already has zeros below — the matrix is now in RREF.

$$x_1 = 1, \quad x_2 = 1, \quad x_3 = 1$$

In reduced form, the last column reads off the solution directly — nothing left to back-substitute.

Free Variables & Parametric Form

Definition

A variable x_i is a free variable if its column, in a row-echelon form of the matrix, is not a pivot column.

$$\begin{bmatrix} 1 & 0 & 5 & 1 \\ 0 & 1 & 2 & -1 \end{bmatrix}$$

Recipe: Parametric Form

1. Write the system as an augmented matrix.
2. Row reduce to reduced row-echelon form.
3. Write the corresponding system of equations.
4. Move all free variables to the right-hand side.

Example — the RREF above translates to a solved system, then the free variable z moves to the right:

$$\begin{cases} x + 5z = 1 \\ y + 2z = -1 \end{cases} \rightarrow \begin{cases} x = 1 - 5z \\ y = -1 - 2z \\ z = z \end{cases} \qquad z \text{ any real number}$$

Source: Margalit & Rabinoff, Interactive Linear Algebra, §1.3, p.24–25

How Many Solutions? Three Possibilities

No Solution

The last (augmented) column is a pivot column.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow 0 = 1$$

Unique Solution

Every non-augmented column is a pivot column.

$$\begin{bmatrix} 1 & 0 & 0 & a \\ 0 & 1 & 0 & b \\ 0 & 0 & 1 & c \end{bmatrix}$$

Infinitely Many

*The last column is not a pivot

*at least one other column is not a pivot column i.e. there is at least 1 free variable

$$\begin{bmatrix} 1 & -2 & 0 & 3 & 1 \\ 0 & 0 & 1 & 4 & -1 \end{bmatrix}$$

Exactly one of these three holds for every linear system — never a fourth option.

Source: Margalit & Rabinoff, Interactive Linear Algebra, §1.3, p.27

For What Values of k...

...does the following system have (a) no solution, (b) exactly one solution, or (c) infinitely many solutions?

$$\begin{cases} kx + y = 1 \\ x + ky = 1 \end{cases}$$

The augmented matrix, and its echelon form:

$$A = \begin{bmatrix} k & 1 & 1 \\ 1 & k & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & k & 1 \\ k & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & k & 1 \\ 0 & 1-k^2 & 1-k \end{bmatrix}$$

$R_1 \leftrightarrow R_2$ $R_2 \rightarrow R_2 - kR_1$

One echelon form, three possible stories — it all depends on the value of $1 - k^2$ in the bottom-right corner.

Three Cases for k

k ≠ ±1 — Unique

Scale row 2 by 1/(1−k²); this is only valid when k² ≠ 1.

$$\begin{bmatrix} 1 & k & 1 \\ 0 & 1 & \dfrac{1}{1+k} \end{bmatrix}, \quad k \neq \pm 1$$

k = −1 — No Solution

Row 2 becomes 0 = 2, which is never true.

$$\begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 2 \end{bmatrix} \rightarrow \; 0 = 2$$

k = 1 — Infinitely Many

Row 2 vanishes; 1 equation in 2 unknowns leaves a free variable.

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \; x_1 + x_2 = 1$$

Same algorithm, same augmented matrix — the parameter just decides which of the three outcomes you land in.

A GEOMETRIC VIEW

The Geometry of Solutions

Every 2×2 system is a pair of lines in the plane. Whether it has one solution, none, or infinitely many is just a question of how those two lines meet.

The General 2×2 Inverse, by Gauss-Jordan

The same [A | I] recipe from Lecture 3 works symbolically, for any invertible 2×2 matrix A:

$$[\,A \mid I\,] \xrightarrow{\text{EROs}} [\,I \mid A^{-1}\,]$$

$$\left[\begin{array}{cc|cc} a & b & 1 & 0 \\ c & d & 0 & 1 \end{array}\right] \rightarrow \left[\begin{array}{cc|cc} 1 & \frac{b}{a} & \frac{1}{a} & 0 \\ 0 & \frac{ad-bc}{a} & -\frac{c}{a} & 1 \end{array}\right] \rightarrow \left[\begin{array}{cc|cc} 1 & 0 & \frac{d}{ad-bc} & -\frac{b}{ad-bc} \\ 0 & 1 & -\frac{c}{ad-bc} & \frac{a}{ad-bc} \end{array}\right]$$

Get a leading 1 in row 1 (÷a), clear column 1 below it, then get a leading 1 in row 2 and clear the rest of column 2 — exactly the same three moves as any Gauss-Jordan reduction.

$$A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \qquad \text{provided } ad - bc \neq 0$$

The Determinant of a 2×2 Matrix

The number ad − bc that appears in every 2×2 inverse has its own name and notation:

$$\det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc = |A|$$

1. The determinant is defined only for square matrices — the determinant of a non-square matrix is not defined.

2. A is invertible if and only if det(A) ≠ 0. In particular, if A^{-1} exists, then det(A) ≠ 0.

det(A) ≠ 0 ⟺ A is invertible ⟺ Ax = b has a unique solution for every b

Source: Anton & Rorres, Elementary Linear Algebra — p.79 (7th Ed.) / p.82 (8th Ed.)

If Ax = b Is a Square System...

$$\det(A) \neq 0$$

A is invertible

There is a unique solution:

$$x = A^{-1}b$$

x = A⁻¹b works for any right-hand side b.

$$\det(A) = 0$$

A⁻¹ doesn't exist

Two possibilities remain:

No solution

Infinitely many

Which one happens is a geometry question — see the next two slides.

One Solution: Intersecting Lines

Consider:

$$\begin{cases} 3x_1 + 4x_2 = 7 \\ x_1 - x_2 = 0 \end{cases}$$

As a matrix equation, with determinant:

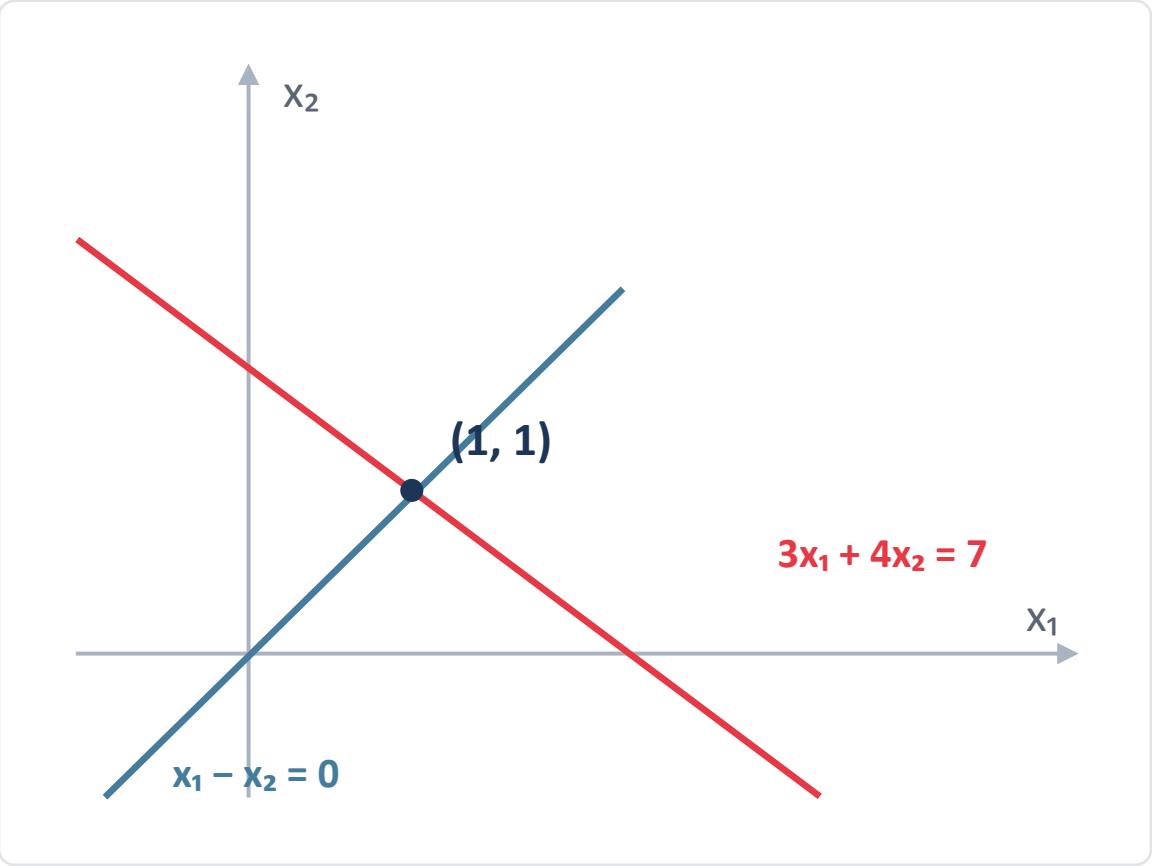
$$\begin{bmatrix} 3 & 4 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 0 \end{bmatrix}$$

$$\det = -3 - 4 = -7 \neq 0$$

Solving $x = A^{-1}b$ gives:

$$(x_1, x_2) = (1, 1)$$

the point where lines ① and ② cross.

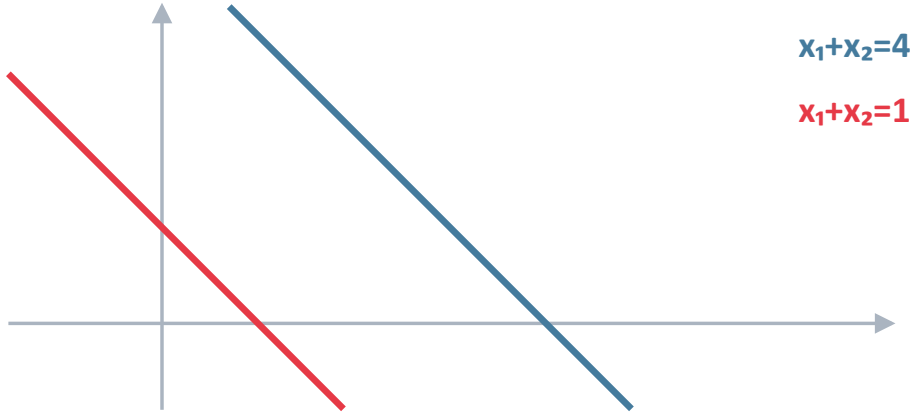


A unique solution is just the (x_1, x_2) point where the two lines cross.

Zero or Infinite: Parallel & Coincident Lines

No Solution — Parallel Lines

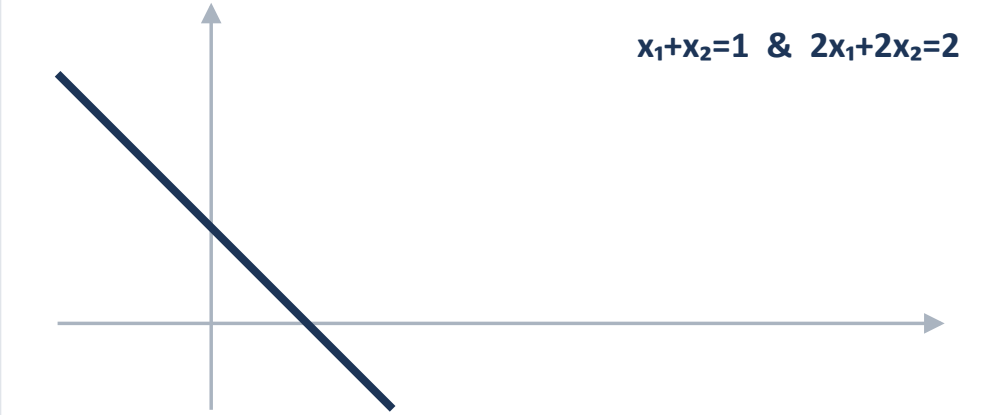
$$\begin{cases} x_1 + x_2 = 4 \\ x_1 + x_2 = 1 \end{cases} \qquad \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 0$$



Same slope, different intercepts — the lines never meet.

Infinitely Many — Coincident Lines

$$\begin{cases} x_1 + x_2 = 1 \\ 2x_1 + 2x_2 = 2 \end{cases} \qquad \begin{vmatrix} 1 & 1 \\ 2 & 2 \end{vmatrix} = 0$$



The second equation is just twice the first — one line, drawn twice.

TRY IT YOURSELF

Practice 1 — RREF Drill

Row reduce this matrix all the way to reduced row-echelon form:

$$\begin{bmatrix} 0 & -2 & 4 & -6 \\ 1 & 1 & 0 & 3 \\ 2 & 0 & 4 & -2 \end{bmatrix}$$

Hints

- No row currently starts with 1 — swap or scale to get one.
- Clear every other entry in that column before moving on.
- Keep going until every pivot column is completely clean, above and below.

Source: Margalit & Rabinoff, Interactive Linear Algebra, §1.3 — pure algorithm practice, no story problem attached.

TRY IT YOURSELF

Practice 2 — For What k?

For what values of k does the following system have (a) no solution, (b) a unique solution, and (c) infinitely many solutions?

$$\begin{cases} kx + 2y = 4 \\ 2x + ky = 4 \end{cases}$$

Hints

- Row reduce the augmented matrix exactly as with today's k example.
- Look for the value(s) of k that make the pivot entry in row 2 vanish.
- Check each such k separately: does it give 0 = (nonzero), or a true but redundant equation?

Same structure as today's lecture, different numbers — confirm you can run the whole classification on your own.

RECAP

Key Takeaways

RREF

clear above and below every pivot — the last column becomes the answer directly.

Free variables

columns without a pivot; move them to the right-hand side to parametrize.

Three outcomes

no solution, one solution, or infinitely many — decided by the pivot pattern.

2×2 inverse

$A^{-1} = 1/\det(A)\ [d\ -b;\ -c\ a]$, provided $\det(A) \neq 0$.

Determinant

ad − bc; defined only for square matrices; A is invertible iff $\det(A) \neq 0$.

Geometry

intersecting lines = unique, parallel = none, coincident = infinitely many.

NEXT TIME

Looking Ahead

Homework

Assignment 2

Posted on both CANVAS and OneNote

Watch out for!

If Practice 2's cases don't match today's pattern, that's the best kind of bug to bring — it almost always means one arithmetic slip in the reduction.

Lecture 5

The Invertible Matrix Theorem

- The master theorem (Anton & Rorres, 9th Ed.): a single square matrix A, and a long list of statements that are all equivalent to “A is invertible.”
- Proving equivalence: showing the statements imply each other around a cycle, not checking every pair directly.
- Special properties of square matrices — this theorem ties together RREF, determinants, and everything else from this chapter.

Questions?

Gauss-Jordan Elimination & the Big Picture · Lecture 4 complete

NEXT: THE INVERTIBLE MATRIX THEOREM